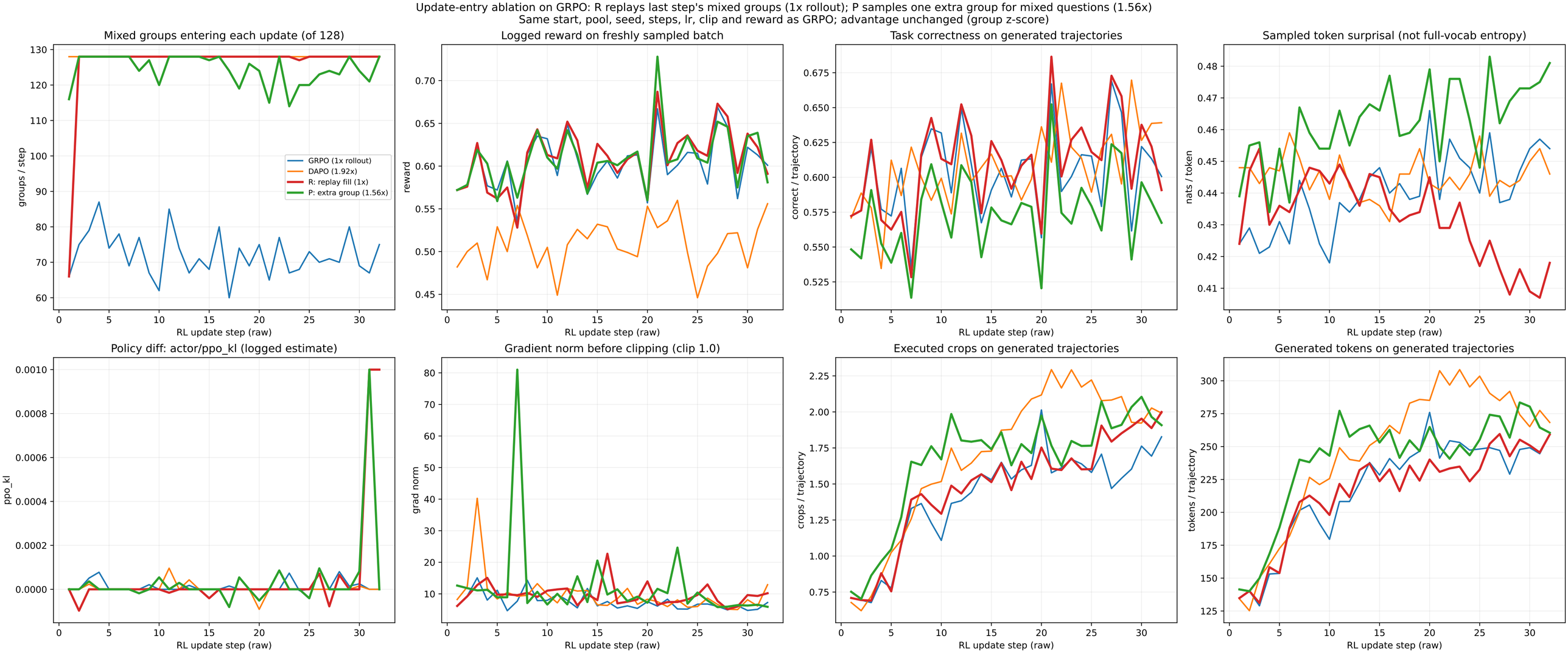
Update-entry ablation on GRPO: R replays last step's mixed groups (1x rollout); P samples one extra group for mixed questions (1.56x)
Same start, pool, seed, steps, lr, clip and reward as GRPO; advantage unchanged (group z-score)
Mixed groups entering each update (of 128)
groups / step
GRPO (1x rollout)
DAPO (1.92x)
R: replay fill (1x)
P: extra group (1.56x)
Logged reward on freshly sampled batch
reward
Task correctness on generated trajectories
correct / trajectory
Sampled token surprisal (not full-vocab entropy)
nats / token
Policy diff: actor/ppo_kl (logged estimate)
ppo_kl
Gradient norm before clipping (clip 1.0)
grad norm
Executed crops on generated trajectories
crops / trajectory
Generated tokens on generated trajectories
tokens / trajectory
RL update step (raw)